Recall

$E = \cdots$

1. Coordinate functionals in a finite dim NLS
2. Norm defined using the coordinate fnals $\|\cdot\|_E$
3. Any finite dim NLS with $\|\cdot\|_E$ norm is a Banach space
4. Finite dim subspace of NLS is closed.
5. Any norms in a F.D space are equivalent
6. Every FD space is a B.S wrt any norm.

: $(C[a,b], \|\cdot\|_\infty)$ is a Banach space.

$(C[a,b], \|\cdot\|_p)$ is a $1 \leq p < \infty$.
is NOT a Banach space.

$\cdot \ (L^p[a,b] - \|\cdot\|_p)$ is a Banach.

Qn: Infinite dimensional space can it be Banach space wrt one norm & NOT a Banach space wrt Another norm.

No.

Results.

1. A Banach space cannot have a denumerable Hamel basis.

2. If a linear space X has a denumerable basis then it cannot be Banach wrt any

a norm.

3. If a l.s is a B.S wrt a norm then it cannot have a denumerable H. B.

Baire Category theorem.

Background.

(Everywhere) dense

$A \subseteq X$ is everywhere dense if $\forall\, x \in X$ and $\epsilon > 0$ $\exists$ a pt $a \in A$ $\ni$ $d(x, a) < \epsilon$

$\Updownarrow$

$B(x, r) \quad \forall r > 0 \quad \forall x \in X$

$B(x,r) \cap A \neq \emptyset$

$\Updownarrow$

Let $x \in X$ $\quad \exists\, a_n \in A$ s.t

$a_n \to x$ as $n \to \infty$.

NOT dense. $\quad A \subseteq X$.

$\exists\, x \in X$ and $\exists\, r > 0$

s.t $B(x, r) \cap A = \emptyset$

$\Rightarrow \overline{A} \neq X$.

NOWHERE Dense

$\forall x \in X \quad \forall B(x, r)$

$$\exists\ B(y, r') \subseteq B(x, r)$$

$$\ni\ B(y, r') \cap A = \emptyset.$$

$$\Updownarrow$$

Ex.

$A \subseteq X$ is nowhere dense if $(\overline{A})^{\circ} = \emptyset$.

All nowhere dense is not dense.

$(0, 1) \subseteq \mathbb{R}$.

Ex.

- Finite set
- Finite union of Finite ~~de~~ set.
- ✗ Countable union of finite sets.

Not dense
$\mathbb{N}$

dense
$\mathbb{Q}$.

1st category set

$S \subseteq X$ is of 1st category if it can be written as a countable union of nowhere dense set.

2nd category

If not then 2nd category.

Baire Category theorem

A complete metric space cannot be of first category.

ie a complete MS cannot be written as a countable union of Nowhere dense set.

<u>Proof</u>: Suppose X is a NLS with denumerable Hamel basis say $E = \{u_1, u_2, \dots\}$

Let $X_n = \text{span}\{u_1, u_2, \dots u_n\}$.

Since X_n is finite dim. so X_n is closed.

- Any proper subspace of a NLS

cannot have any int point.

$A \underset{\neq}{\subset} X$ say $y \in \text{int}(A) = A^0$

$\exists\ r > 0$ s.t $B(x, r) \subseteq A$.

Let $y \in X$

$$z = x + \frac{r}{2}\frac{y}{\|y\|}$$

$$\|z - x\| = \frac{r}{2} < r$$

$$\Rightarrow z \in B(x, r) \subseteq A$$

$\therefore\ x \in A,\ z \in A.$, A proper subspace.

$$y = \frac{2\|y\|}{r}(z - x) \in A.$$

$\therefore$ $\text{int}(A) = \phi$

• $X_n \to$ closed & $X_n \subseteq X$

$$\overline{X_n}^{\circ} = X_n^{\circ} = \phi$$

$\therefore$ X_n's are nowhere dense sets

& since $X = \bigcup_n X_n \Rightarrow X$ is not complete (By Baire category theorem).

Consequences.

1. $\left(\mathcal{P}[a,b], \|\cdot\|_\infty\right)$ not a Banach space.

Denumerable basis

$E = \{u_1, \ldots \}$

$$u_j(t) = t^{j-1}$$

2. c_{00} has a denumerable Hamel basis

$$\{ e_1, e_2, \cdots \}$$

$$e_j = (0, 0, \ldots, \underset{j\text{th}}{1}, 0, \cdots 0).$$

If $x \in c_{00}$

$$x = (x(1), x(2), \ldots)$$

$$= x(1) e_1 + x(2) e_2 + \cdots$$

c_{00} is NOT a Banach space.

$x \in \ell^p$.

$x = (x(1), x(2) - - \quad)$

$\sum |x(j)|^p < \infty$

$\{e_1, e_2, \ldots\}$ is not a Hamel basis for ℓ^p.

$p = 1$

$x = (1, \frac{1}{2}, \frac{1}{2^2}, \ldots) \in \ell^1$

$\sum \frac{1}{2^n} < \infty$

$x \neq$ l.c of finitely many $\{e_1, e_2, \ldots\}$.

If X is a Banach space

i) Either X is finite dim.

2. X has a basis consisting of uncountable number of elements.

Schauder Basis.

A seq $\{u_n\}$ in a NLS is said to be a Schauder Basis of X if

$\forall x \in X$ $\exists!$ (α_n), $\alpha_n \in \mathbb{K}$.

s.t
$$x = \sum_{j=1}^{\infty} \alpha_j u_j \quad \text{ie}$$

$$S_N = \sum_{j=1}^{N} \alpha_j u_j \Rightarrow \|S_N - x\| \to 0$$

as $N \to \infty$.

$$\left[X = \text{span}\{u_1 \ldots u_n\} \quad \overline{\text{span}\{u_1 \ldots u_n \ldots\}} = X \right]$$

Remark:

1. Schauder basis if exists then it is LI.

$$\sum_{i=1}^{n} \beta_i u_i = 0$$

Let $\beta_j = 0$, $j > n$

Then

$$\sum_{j=1}^{\infty} \beta_j u_j = 0 = \sum_{j=1}^{\infty} 0 \cdot u_j$$

$$\Rightarrow \beta_j = 0 \quad \forall j \in \mathbb{N}.$$

$\{u_1, u_2 - - - \}$ are lin inp.

(2) Then $x \in X$,

$$x = \sum_{j=1}^{\infty} \alpha_j u_j$$

$$= \lim_{n \to \infty} \sum_{j=1}^{n} \alpha_j u_j = \lim_{n \to \infty} y_n$$

$$y_n = \sum_{j=1}^{n} \alpha_j u_j$$

$\in$ span $\{u_1 \cdots u_n\}$

$$\Rightarrow x \in \overline{\text{span}\{u_1, \ldots u_n, \cdots\}}$$

$\therefore$ span$\{u_1, \dots u_n, \dots\}$ is dense in X.

③ Let X possess a Hamel basis say

$E = \{u_1, \dots \}$.

Then E is a Schauder basis.

Ex.

1.) $(C_0, \|\cdot\|_\infty)$

$E = \{e_1 \dots e_n, \dots\} \rightarrow$ Hamel basis

$\Downarrow$

Schauder basis.

2. $(P[a,b], \|\cdot\|_\infty)$

$u_j(t) = t^{j-1}$

3. $\left(\ell^p, \|\cdot\|_p\right)$

$E = \{e_1, e_2, \ldots\}$ $\quad x \in X$

$x = (x(1), x(2) \ldots)$

$$\sum_{j=1}^{n} x(j) e_j = (x(1), x(2), x(3), \ldots x(n), 0 \ldots)$$

$$\|x - \sum_{j=1}^{n} x(j) e_j\|_p$$

$$= \|(x(1), x(2), x(3) \ldots x(n), x(n+1) \ldots)$$

$$- (x(1), x(2), \ldots x(n), 0, \ldots 0)\|_p$$

$$= \|(0, 0 \ldots 0, x(n+1), \ldots)\|_p$$

$$= \sum_{j=n+1}^{\infty} |x(j)|^p < \epsilon \quad \forall n \geq N.$$

$$\therefore \quad \| x - \sum_{j=1}^{n} x(j) e_j \|_p \to 0 \quad \text{as } n \to \infty$$

Uniqueness ??

$\{e_1, e_2, \ldots e_n, \ldots\} \to$ Schauder basis.

---- x ----

$(C[0,1], \|\cdot\|_\infty)$ — Faber Schauder system

— Franklin system.

Seperability

$A \subseteq X$, $A \to$ dense, $A \to$ countable.

Ex. • $X = \mathbb{R}$.

$A = \mathbb{Q}$.

• $X = \mathbb{C}$ $\quad A = \mathbb{Q} + i\mathbb{Q}$

Ex. • Every subspace of a sep. m.s is seperable.

• $A \underset{\text{dense}}{\subseteq} X$, A is seperable

$\Rightarrow X$ is seperable.

Theo. Every finite dimensional NLS is separable.

Proof. Let X be a f.d NLS.
$E = \{u_1, \dots, u_n\}$ be a basis for X.

For each $x \in X$

$$x = \sum_{j=1}^{n} \alpha_j u_j$$

$\alpha_j \in \mathbb{K}$.

$$\mathbb{Q}_K = \begin{cases} \mathbb{Q}, & \mathbb{K} = \mathbb{R} \\ \mathbb{Q} + i\mathbb{Q}, & \mathbb{K} = \mathbb{C} \end{cases}$$

$$D = \left\{ \sum_{j=1}^{n} \beta_j u_j \;\middle|\; \beta_j \in \mathbb{Q}_K \right\}.$$

- D is countable.
- D dense (?)

$$x = \sum_{j=1}^{n} \alpha_j u_j$$

By denseness of $\mathbb{Q}_K$ for each j, & α_j $\exists$ β_j s.t

$$|\alpha_j - \beta_j| < \epsilon/2 \quad \forall j = 1, 2 \ldots n$$

$$\left\| x - \sum_{j=1}^{n} \beta_j u_j \right\| = \left\| \sum_1^n \alpha_j u_j - \sum_1^n \beta_j u_j \right\|$$

$$\leq \sum_{j=1}^{n} |\alpha_j - \beta_j| \|u_j\|$$

$$< \epsilon \sum_{j=1}^{n} \|g u_j\|$$

$\therefore$ D is dense in X.